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**A TRANSFER LEARNING-BASED MOBILE APPLICATION FOR NAIL
DISEASE DETECTION**

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DECLARATION

This is certifying that the thesis titled “**A Transfer Learning-Based Mobile Application for Nail Disease Detection**” is our own research carried out under the guidance of our supervisor of Mohammad Arif Hasan Chowdhury Assistant Professor, Department of Computer Science and Engineering, University of Science and Technology Chittagong. This thesis has not been submitted, either in whole or in part, for the award of any degree or diploma at any other university or institution.

All sources of information and materials used in this study have been properly acknowledged through appropriate references. Any assistance received in the course of this research has been duly acknowledged. The results, analysis, and conclusions presented in this thesis are original and based on the experimental work conducted by us.

We further declare that no part of this thesis has been copied from any other source without proper citation and that the work complies with the academic integrity and ethical standards of the university.

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APPROVAL CERTIFICATE

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DEDICATION

This project is dedicated to all the parents.

ABSTRACT

Nail abnormalities serve as important clinical indicators of both localized dermatological conditions and underlying systemic disorders. However, access to specialized dermatological care remains limited in many developing regions, leading to delayed diagnosis and treatment. This thesis presents the design and

development of an AI-powered mobile application for automated nail disease detection using transfer learning techniques. The proposed system performs multi-class classification of four nail conditions: Onychomycosis (fungal infection), Nail Psoriasis, Clubbing Nail, and Healthy Nail.

Through systematic experimentation and comparative analysis of multiple deep learning architectures including VGG16, VGG19, MobileNetV1/V2, DenseNet121, and EfficientNetB0, we identified EfficientNetB0 as the optimal backbone achieving the best balance between accuracy (86.25% test accuracy) and computational efficiency (5.3 million parameters). The proposed methodology incorporates a two-phase transfer learning strategy, resolution upscaling to 260×260 pixels, class weighting to address data imbalance, L2 regularization, and test-time augmentation to enhance model robustness.

The trained model is deployed through a FastAPI-based cloud backend and integrated with a cross-platform mobile application developed using React Native. The complete system enables users to capture nail images, receive real-time predictions with confidence scores, access disease information, and generate downloadable PDF reports for documentation and sharing.

Experimental results demonstrate strong class-wise performance with precision scores of 1.00 for Clubbing Nail and 0.94 for Onychomycosis. Qualitative error analysis reveals that misclassifications primarily occur due to limitations in 2D image acquisition (top-down views missing 3D curvature for clubbing detection) and visual similarities between early-stage disease manifestations. The findings confirm that transfer learning-based approaches can effectively support preliminary nail disease screening in resource-constrained environments, though the system is intended as a decision-support tool rather than a replacement for professional medical diagnosis.

Keywords: Transfer Learning, Nail Disease Detection, EfficientNet, Deep Learning, Mobile Healthcare, Onychomycosis, Nail Psoriasis, Clubbing Nail.

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CHAPTER 1

INTRODUCTION

Human nails are not merely protective structures of the distal phalanx; they are clinically significant biological indicators that often reflect both localized dermatological conditions and underlying systemic disorders. Medical research has long established that abnormalities in nail color, texture, thickness, curvature, and surface morphology can serve as early visual markers for a wide range of diseases, including fungal infections, autoimmune disorders, cardiovascular abnormalities, and respiratory illnesses [10], [13].

Among common nail-related disorders, onychomycosis, nail psoriasis, and clubbing are particularly important due to their strong association with systemic health conditions. Onychomycosis, a fungal infection of the nail plate, affects a significant portion of the global population and often leads to nail discoloration, thickening, and crumbling. Nail psoriasis, an autoimmune manifestation, presents distinct visual markers such as pitting, oil-drop discoloration, and onycholysis. Nail clubbing, characterized by bulbous enlargement of the distal digits and an increased Lovibond angle, is frequently associated with chronic pulmonary, cardiac, and gastrointestinal diseases [3], [4], [13].

Despite their diagnostic importance, nail diseases are frequently underdiagnosed or misinterpreted, particularly in developing countries. Many individuals perceive nail abnormalities as cosmetic concerns rather than medical symptoms, leading to delayed diagnosis and treatment. This issue is further aggravated in rural and resource-limited regions, where access to dermatologists, laboratory facilities, and specialized diagnostic tools is limited [6], [12].

Conventional nail disease diagnosis primarily relies on clinical inspection followed by laboratory-based confirmation techniques such as potassium hydroxide (KOH) tests, fungal culture, microscopic examination, or biopsy procedures. Although these methods are clinically reliable, they are time-consuming, expensive, and require trained personnel and laboratory infrastructure. Consequently, such diagnostic approaches are not suitable for large-scale screening or early-stage disease detection, especially in underserved populations [10].

To address these limitations, early research efforts explored automated nail disease detection using traditional image processing techniques. These systems extracted handcrafted visual features—such as color histograms, texture descriptors, and shape features—from nail images and applied classical machine learning classifiers for disease identification. While these approaches demonstrated feasibility under controlled conditions, their performance degraded significantly in real-world environments due to sensitivity to illumination variation, background clutter, camera quality, and finger orientation [10], [13].

The evolution of machine learning introduced data-driven classification approaches, enabling improved diagnostic performance compared to rule-based systems. Algorithms such as K-Nearest Neighbors (KNN), Support Vector Machines (SVM), and Random Forest were applied to handcrafted features extracted from nail images. Although these models improved

accuracy, they still relied heavily on manual feature engineering and domain expertise, limiting scalability and adaptability across diverse datasets [10].

Recent advances in deep learning, particularly Convolutional Neural Networks (CNNs), have transformed medical image analysis by enabling automatic hierarchical feature extraction directly from raw images. CNN-based models have demonstrated superior performance in capturing subtle visual patterns, making them highly suitable for nail disease classification tasks [4]. However, training deep neural networks from scratch requires large, well-annotated datasets, which are scarce in nail disease research.

To overcome data scarcity, transfer learning has emerged as a dominant strategy. By leveraging pre-trained models such as VGG, MobileNet, DenseNet, and EfficientNet, researchers have successfully transferred knowledge learned from large-scale datasets to nail disease classification problems, achieving improved accuracy and faster convergence with limited data [2], [8], [9].

Parallel to algorithmic advancements, the rapid proliferation of smartphones has enabled the development of mobile healthcare (mHealth) solutions. Smartphones equipped with high-resolution cameras and internet connectivity provide a practical platform for image-based disease screening. Mobile applications integrated with AI-based diagnostic systems offer real-time feedback, preliminary health assessment, and educational guidance, significantly improving healthcare accessibility in underserved regions [6], [12].

However, deploying deep learning models in mobile environments introduces new challenges related to computational efficiency, memory constraints, inference latency, and system integration. Lightweight architectures optimized for mobile deployment, such as MobileNet and EfficientNet, have therefore gained prominence due to their balance between accuracy and efficiency [6], [11].

Motivated by these developments, this research focuses on designing an AI-driven, mobile-friendly nail disease detection system that integrates optimized transfer learning models with a complete end-to-end pipeline, including image acquisition, preprocessing, automated classification, and report generation. The proposed system aims to support early-stage screening, improve diagnostic accessibility, and enhance health awareness, particularly in rural and resource-constrained environments.

1.1 BACKGROUND AND MOTIVATION

Nails play a significant role in reflecting an individual's overall health condition. Variations in nail color, texture, thickness, and shape are often associated with a wide range of dermatological and systemic diseases. Conditions such as onychomycosis, psoriasis, and nail clubbing may indicate fungal infections, autoimmune disorders, or underlying respiratory and cardiovascular abnormalities. As a result, nail examination has long been recognized as a valuable, non-invasive diagnostic indicator in clinical practice.

Despite their diagnostic importance, nail diseases are frequently overlooked, particularly in developing countries. Many individuals tend to ignore early nail abnormalities due to lack of awareness, social stigma, or the perception that nail disorders are merely cosmetic issues. In rural and resource-limited regions, access to dermatologists and specialized diagnostic facilities is often inadequate, leading to delayed diagnosis and improper treatment [10], [13]. Such delays can cause disease progression, recurrence, and increased healthcare burden.

Traditional nail disease diagnosis relies on clinical visual inspection, laboratory-based fungal culture tests, microscopic examination, or biopsy procedures. While these methods are clinically reliable, they are time-consuming, costly, and require trained medical professionals and laboratory infrastructure. These constraints limit their availability and practicality for large-scale screening or early-stage detection, especially in underserved communities [10].

Early research efforts attempted to automate nail disease detection using traditional image processing techniques. These approaches focused on handcrafted feature extraction methods such as color histogram analysis, texture descriptors, and edge detection to identify nail abnormalities. Although these methods provided initial insights into nail image analysis, they were highly sensitive to illumination changes, background noise, camera quality, and finger positioning [10], [13]. Consequently, their performance degraded significantly in real-world scenarios involving smartphone-acquired images.

The emergence of machine learning introduced data-driven approaches for nail disease classification. Classical classifiers such as KNN, SVM, and Random Forest were applied to handcrafted features to improve diagnostic accuracy. However, these approaches still depended heavily on manual feature engineering and domain expertise, limiting their scalability and adaptability to diverse datasets [10].

Recent advances in deep learning, particularly convolutional neural networks (CNNs), have revolutionized medical image analysis. CNN-based models automatically learn hierarchical feature representations directly from raw images and have demonstrated superior performance in detecting complex visual patterns. In the context of nail disease detection, deep learning approaches have achieved significantly higher accuracy compared to traditional methods by effectively capturing subtle texture and color variations [4].

However, training deep neural networks from scratch requires large annotated datasets, which are scarce in nail disease research. To overcome this challenge, transfer learning has become a widely adopted strategy. By utilizing pre-trained models such as VGG, MobileNet, and EfficientNet, researchers have successfully transferred learned visual representations from large-scale datasets to nail disease classification tasks, achieving improved accuracy and faster convergence with limited data [2], [8].

In parallel, the rapid growth of smartphone technology has enabled the development of mobile-based healthcare applications. Smartphones equipped with high-resolution cameras and internet connectivity provide an ideal platform for image-based disease screening. Mobile healthcare applications integrated with AI-based diagnostic systems offer real-time feedback,

preliminary health assessment, and educational guidance, reducing dependency on immediate clinical visits [6], [12].

Lightweight deep learning architectures optimized for efficiency have further facilitated mobile deployment. Models such as MobileNet and EfficientNet balance classification accuracy with computational efficiency, making them suitable for real-time inference in mobile healthcare environments [6], [11]. Cloud-based backend architectures also enable resource-intensive models to be deployed without overloading mobile devices.

1.2 PROBLEM STATEMENT

Despite the diagnostic significance of nail conditions, several challenges limit the effectiveness and accessibility of nail disease diagnosis. The key problems addressed in this research are outlined as follows:

1. Limited Access to Specialized Healthcare: In many developing and rural regions, access to dermatologists and diagnostic facilities is limited. As a result, nail diseases are often overlooked, misdiagnosed, or treated at advanced stages, leading to disease progression and recurrence [10], [13].

2. Limitations of Conventional Diagnostic Methods: Traditional diagnostic techniques such as clinical examination, fungal culture tests, and microscopic analysis are time-consuming, costly, and require trained medical professionals and laboratory infrastructure. These constraints make them unsuitable for large-scale screening and early-stage detection in resource-constrained environments [10].

3. Poor Generalization of Traditional Image Processing Methods: Early automated nail disease detection systems based on handcrafted image features are highly sensitive to illumination variation, background noise, camera quality, and finger orientation. Consequently, these methods perform poorly on real-world smartphone-acquired images [10], [13].

4. Computational Constraints of Existing Deep Learning Models: Although deep learning approaches have demonstrated improved accuracy in nail disease classification, many existing models are computationally expensive and require large annotated datasets. Such models are not optimized for real-time or mobile deployment, limiting their practical usability [4], [6].

5. Lack of Integrated End-to-End Mobile Diagnostic Systems: Most existing studies focus on standalone classification models without providing a complete system that integrates image acquisition, preprocessing, automated analysis, result interpretation, and user-friendly reporting. The absence of integrated mobile solutions reduces adoption by non-expert users and limits healthcare accessibility [6], [12].

1.3 OBJECTIVES OF THE STUDY

The primary objectives of this research are as follows:

- To design and optimize a transfer learning–based deep learning framework for robust multi-class nail disease classification.
- To develop an end-to-end mobile-integrated diagnostic system enabling real-time nail image analysis and automated report generation.
- To evaluate and validate the clinical feasibility and deployment efficiency of the proposed system for rural and resource-constrained environments.

1.4 SCOPE OF THE STUDY

This study focuses on developing an AI-driven mobile application for teledermatology. The goal is to create a system that can automatically identify nail diseases using photos taken on a smartphone. This research explores the technical hurdles of making advanced AI models work smoothly in mobile-first healthcare settings, especially where resources are limited.

1.4.1 TECHNICAL SCOPE

The core technical work of this project involves several key layers, from the AI "brain" to the user interface:

Deep Learning Architecture: We designed a specialized AI framework using EfficientNetB0. This model uses advanced techniques like compound scaling and two-phase fine-tuning to ensure it can accurately identify diseases from different types of photos.

Image Processing Pipeline: To help the AI see better, we built a workflow that resizes images to a high-quality 260×260 pixel format and "cleans" them (normalization). This ensures that the model focuses on the right features regardless of lighting or background.

Multi-Class Classification: The system is trained to recognize four specific categories:

- Healthy nails
- Onychomycosis (fungal infections)
- Nail psoriasis
- Clubbing (often a sign of internal health issues)

Backend & Mobile Development: We built a Flask-based API to handle the heavy AI calculations on a server, allowing the mobile app (built with React Native) to remain fast and lightweight.

Automated Reporting: The app includes a system that generates a PDF report. This report gives users their results, how confident the AI is in its prediction, and educational tips for their next steps.

1.4.2 LIMITATIONS AND BOUNDARIES

While this system is a powerful tool, it has specific boundaries:

Disease Coverage: The study is restricted to the four conditions mentioned above. It cannot currently detect other issues like nail melanoma (cancer) or lichen planus due to the lack of available data for those conditions.

Dataset and 2D Constraints: The AI was trained on a limited set of images. Because it uses 2D photos, it might miss some "3D" details (like the specific angle of a curved nail) that a doctor could see in person.

Connectivity: The app currently needs an internet connection to work because the AI "brain" lives on a remote server rather than directly on the phone.

Not a Final Diagnosis: This tool is designed for preliminary screening. It is meant to support decisions, not replace a professional doctor's diagnosis or a clinical lab test.

Intended Use Case

The system is designed to be a helpful resource in the following scenarios:

Rural Healthcare: Providing a "digital specialist" in areas where skin doctors are hard to find.

Self-Assessment: Helping individuals stay aware of their health and catch potential issues early.

Education: Serving as a learning tool for medical students and healthcare workers to practice identifying nail conditions.

Triage Support: Helping primary care clinics decide which patients need to see a specialist most urgently.

CHAPTER 2

LITERATURE REVIEW

2.1 NAIL DISEASE DETECTION USING IMAGE PROCESSING TECHNIQUES

Early research on automated nail disease detection mainly focused on traditional image processing techniques. These methods aimed to analyze visible characteristics of nails, such as color variation, texture patterns, and structural changes, to identify abnormalities associated with different diseases. Due to their low computational requirements, these approaches were initially considered suitable for basic diagnostic support systems.

Several studies utilized color-based feature extraction techniques, particularly RGB and grayscale intensity analysis, to differentiate between healthy and diseased nails. Variations in nail color were used as indicators of conditions such as fungal infections and systemic diseases. Early disease detection systems extracted handcrafted features such as color histograms and texture descriptors and applied classical classifiers for nail disease identification [10], [13], [1]. Although these systems demonstrated reasonable accuracy under controlled environments, their performance degraded significantly in real-world smartphone images due to illumination and background variation [10], [13], [8].

Texture-based analysis was also explored to identify surface irregularities, ridges, and nail plate deformities. Structural features extracted from nail images were used to detect early disease symptoms related to systemic health conditions [13]. However, these approaches were highly sensitive to illumination changes, shadows, and camera angle variations, which limited their robustness.

Some studies attempted to improve detection accuracy by combining multiple handcrafted features, including color histograms, edge detection, and texture descriptors. While such feature fusion techniques enhanced classification performance to some extent, they required careful manual tuning and expert knowledge, reducing scalability and adaptability to new datasets [10], [13].

More recent works introduced convolutional neural networks while still relying partially on conventional image preprocessing and enhancement techniques. These studies demonstrated that basic image processing steps such as contrast enhancement and segmentation could highlight nail abnormalities, but standalone image processing techniques were insufficient for reliable disease classification [1].

A major limitation identified across traditional image processing-based methods is their poor generalization capability. Nail images captured using smartphones exhibit significant variations in lighting conditions, background clutter, skin tone, and image resolution. Handcrafted feature-based systems struggle to handle such variability, resulting in inconsistent performance and high misclassification rates [10].

2.2 MACHINE LEARNING-BASED APPROACHES FOR NAIL DISEASE CLASSIFICATION

With the limitations of traditional image processing techniques, researchers began adopting machine learning algorithms to improve nail disease classification performance. These approaches rely on handcrafted feature extraction followed by supervised learning classifiers to identify disease patterns from nail images.

Several studies applied classical machine learning classifiers such as Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Artificial Neural Networks (ANN), and Random Forest (RF) for nail disease detection. Feature sets typically included color histograms, texture descriptors, statistical measures, and shape-based attributes. Early disease diagnosis systems demonstrated that machine learning classifiers could outperform rule-based image processing methods when trained on well-defined feature representations [10].

Ensemble learning techniques were also explored to improve classification robustness. An ensemble-based KNN framework was used to enhance early disease detection from nail images by aggregating predictions from multiple feature representations [10]. Although ensemble methods improved classification accuracy, their performance remained highly dependent on feature quality and consistency.

Some studies investigated hybrid machine learning approaches combining deep feature extraction with classical classifiers. For example, features extracted using convolutional neural networks were classified using Random Forest or Logistic Regression classifiers to improve performance on limited datasets [2], [8]. These hybrid strategies demonstrated that machine learning classifiers can effectively utilize deep features when end-to-end deep learning training is constrained by data availability.

Despite improved accuracy compared to pure image processing methods, machine learning-based nail disease detection systems suffer from several limitations. The reliance on handcrafted feature engineering requires domain expertise and extensive experimentation. Moreover, handcrafted features often fail to capture complex visual patterns present in nail diseases, particularly when disease manifestations are subtle or visually similar.

Another challenge is the scalability of machine learning models. As dataset size and disease categories increase, manual feature extraction becomes inefficient and less adaptable. Additionally, many machine learning approaches lack robustness against variations in illumination, background clutter, and camera quality commonly observed in smartphone-acquired images [13].

2.3 DEEP LEARNING AND CNN-BASED NAIL DISEASE DETECTION

The introduction of deep learning, particularly Convolutional Neural Networks (CNNs), significantly advanced the field of nail disease detection by enabling automatic feature extraction from raw images. Unlike traditional machine learning approaches, CNN-based

methods eliminate the need for handcrafted features and are capable of learning complex spatial and texture patterns directly from nail images.

Several studies demonstrated that CNN-based architectures outperform classical machine learning classifiers in nail disease classification tasks. Deep learning models were applied to detect multiple nail disorders by learning discriminative visual features such as discoloration, nail plate deformation, and surface texture variations [4]. These approaches showed improved robustness against image noise and background variability compared to earlier methods.

Transfer learning-based CNN architectures were also explored to address the limited availability of labeled nail image datasets. Pre-trained networks such as VGG16, VGG19, and MobileNet were fine-tuned for nail disease classification, resulting in improved accuracy and faster convergence [8]. These studies highlighted the effectiveness of reusing knowledge from large-scale image datasets for medical image analysis.

Lightweight CNN architectures gained attention due to their suitability for mobile and real-time applications. Mobile-optimized models were used to reduce computational complexity while maintaining acceptable classification performance [6]. However, some lightweight models exhibited reduced accuracy when distinguishing between visually similar nail diseases.

Recent research also explored advanced deep learning architectures to further enhance performance. Capsule Network-based models integrated with MobileNet backbones demonstrated improved feature representation and spatial relationship modeling for nail disease classification [11]. These models achieved high accuracy but increased architectural complexity.

Real-time detection approaches were introduced using object detection frameworks such as YOLO. These systems enabled rapid localization and classification of nail disease regions in images and videos [12]. While effective for real-time analysis, such models often require extensive computational resources and large annotated datasets.

Despite their strong performance, CNN-based nail disease detection systems face challenges related to overfitting, dataset imbalance, and deployment constraints. Many deep models are computationally intensive and not optimized for mobile healthcare environments. These limitations highlight the need for efficient architectures and deployment-aware system design.

2.4 TRANSFER LEARNING FOR NAIL DISEASE DETECTION

Transfer learning has emerged as a powerful paradigm in medical image classification, particularly in domains where large-scale labeled datasets are unavailable. Instead of training a deep neural network from scratch, transfer learning reuses knowledge acquired from a source domain and applies it to a related target domain. Formally, given a source task (T_s) and a target task (T_t), transfer learning aims to improve the learning of (T_t) using knowledge from (T_s), especially when the target dataset is limited.

In medical image analysis, large annotated datasets are difficult to obtain due to privacy concerns, clinical validation requirements, and expert labeling constraints. To address this limitation, pre-trained convolutional neural networks trained on large-scale datasets such as ImageNet have been widely adopted. These models learn generic low-level and mid-level visual features, including edges, textures, shapes, and spatial hierarchies, which can be effectively transferred to medical imaging tasks.

Two primary transfer learning strategies are commonly used:

1) Feature Extraction Approach: The convolutional base of a pre-trained network remains frozen, and only the final classification layers are retrained. This approach reduces computational cost and prevents overfitting in small datasets.

2) Fine-Tuning Approach: Selected deeper layers of the network are unfrozen and retrained along with the classification head. This allows adaptation of high-level semantic features to domain-specific patterns, such as nail pathology characteristics.

Several studies applied transfer learning for nail disease classification using pre-trained architectures such as VGG16, VGG19, and MobileNet [2], [8]. These models demonstrated improved convergence speed and classification accuracy compared to CNNs trained from scratch. However, deeper networks such as VGG often suffer from overfitting under limited dataset conditions due to their large parameter size.

Lightweight architectures, particularly MobileNet-based models, have been explored to reduce computational complexity and enable mobile deployment [6]. While these models provide efficiency benefits, they may struggle to capture both global geometric patterns and fine-grained texture features simultaneously.

Recent studies highlighted the effectiveness of EfficientNet-based architectures for medical image classification tasks due to their compound scaling mechanism [9]. By uniformly scaling network depth, width, and resolution, EfficientNet achieves improved accuracy while maintaining parameter efficiency. This makes it particularly suitable for mobile healthcare applications where computational resources are constrained.

Hybrid transfer learning approaches have also been proposed, combining CNN-based feature extraction with classical classifiers such as Random Forest [2]. Although such models enhance robustness, they introduce additional architectural complexity and may increase inference time.

Despite the demonstrated advantages of transfer learning, challenges remain. Improper fine-tuning depth, aggressive learning rates, and dataset imbalance can degrade pre-trained feature representations. Additionally, many existing studies focus solely on accuracy improvement without considering deployment efficiency in real-world mobile healthcare environments.

2.5 HYBRID DEEP LEARNING MODELS

Hybrid deep learning models have gained significant attention in nail disease detection due to their ability to combine the powerful feature extraction capability of deep neural networks with the robustness of traditional machine learning classifiers. These models aim to overcome the limitations of both standalone deep learning and classical machine learning approaches, especially when training data is limited.

Several studies demonstrated that convolutional neural networks can be effectively used as feature extractors, while machine learning classifiers such as Random Forest, Support Vector Machine, or Logistic Regression are employed for final classification. This hybrid strategy reduces overfitting and improves generalization performance by leveraging ensemble-based decision mechanisms [2].

Hybrid architectures using VGG16 as a feature extractor followed by a Random Forest classifier achieved high classification accuracy in fingernail disease detection tasks [2]. These models outperformed conventional end-to-end CNN classifiers, particularly in scenarios where dataset size was insufficient for deep network fine-tuning. The results indicate that Random Forest classifiers can effectively exploit deep feature representations for robust decision-making.

Similar hybrid strategies were explored in multi-disease classification frameworks, where deep features extracted from transfer learning models were classified using classical algorithms to enhance stability and accuracy [8]. These approaches demonstrated improved performance compared to pure CNN-based classifiers, especially for visually similar disease classes.

Beyond nail disease analysis, hybrid deep learning models have also been successfully applied in related medical imaging domains, reinforcing their suitability for biomedical classification tasks [5]. Comparative studies showed that transfer learning-based hybrid models outperform shallow CNN architectures in terms of accuracy and convergence speed, particularly when training data is scarce.

Despite their advantages, hybrid models introduce additional design complexity. Selecting appropriate feature layers, classifier types, and hyperparameters requires careful experimentation. Moreover, hybrid architectures may increase inference time due to the inclusion of separate classification stages.

2.6 NAIL DISEASE DATASETS AND CHALLENGES

The availability and quality of datasets play a crucial role in the performance of automated nail disease detection systems. One of the most significant challenges identified in existing research is the lack of large, standardized, and publicly available nail image datasets. Most studies rely on small, manually collected datasets obtained from clinical sources or publicly shared repositories, which limits model generalization [4], [8].

Class imbalance is a common issue in nail disease datasets, where certain disease categories are underrepresented compared to healthy nail images. This imbalance often leads to biased model predictions favoring majority classes, thereby reducing diagnostic reliability [8]. Several studies attempted to address this problem using class weighting and data augmentation techniques; however, imbalance remains a persistent challenge.

Image variability further complicates dataset quality. Nail images captured using smartphones exhibit significant variations in lighting conditions, camera resolution, skin tone, background clutter, and finger orientation. Such inconsistencies introduce noise into the training data and negatively affect model performance [10], [13].

Another challenge lies in inconsistent annotation quality. Many datasets rely on visual labeling without laboratory confirmation, increasing the risk of mislabeling. This issue directly impacts supervised learning models, as incorrect labels can degrade classification accuracy [4].

To mitigate these challenges, researchers emphasized the importance of image preprocessing and augmentation strategies. Techniques such as resizing, contrast enhancement, noise reduction, rotation, and flipping were applied to improve dataset diversity and robustness [8]. While these methods enhance generalization to some extent, they cannot fully compensate for limited dataset size.

2.7 MOBILE AND HEALTHCARE-ORIENTED APPLICATIONS

Mobile healthcare applications have become increasingly important in improving accessibility to medical diagnostic services, particularly in rural and underserved regions. The widespread availability of smartphones equipped with high-resolution cameras has enabled the development of mobile-based disease screening systems that support early detection and health awareness.

Several studies explored mobile-based diagnostic systems using convolutional neural networks for medical image analysis. Mobile-friendly deep learning architectures were applied to enable on-device or server-based inference while maintaining acceptable accuracy [6]. These approaches demonstrated the feasibility of integrating AI-powered diagnosis into mobile platforms.

In the context of nail disease detection, mobile applications were developed to capture nail images, perform automated analysis, and provide diagnostic feedback to users. Such systems emphasized usability and real-time performance, allowing non-expert users to obtain preliminary health assessments without visiting specialized clinics [4].

Lightweight architectures such as MobileNet and EfficientNet were frequently adopted due to their reduced computational complexity and suitability for deployment in mobile environments [6], [11]. These models achieved a balance between accuracy and efficiency, making them suitable for real-time healthcare applications.

Cloud-based inference architectures were also explored, where mobile applications communicate with backend servers through RESTful APIs. This design allows computationally intensive models to run on remote servers while keeping the mobile application lightweight [12]. However, such systems depend on reliable internet connectivity, which may limit usability in low-connectivity regions.

2.8 FUNGAL NAIL DISEASES AND RELATED STUDIES

Fungal nail diseases, particularly onychomycosis, represent one of the most common nail disorders worldwide and account for a significant portion of nail-related clinical cases. These infections are caused by dermatophytes, yeasts, or non-dermatophyte molds and often lead to nail discoloration, thickening, brittleness, and deformation. Early detection is essential, as delayed diagnosis can result in chronic infection and recurrence.

Several studies focused on detecting fungal nail infections using image-based analysis. Deep learning-based approaches demonstrated strong capability in identifying visual characteristics associated with fungal infections, such as yellowish discoloration, nail plate thickening, and irregular surface texture [4]. These models showed improved accuracy compared to traditional methods, particularly when trained using transfer learning techniques.

Transfer learning-based multi-class classification frameworks included fungal nail disease as a primary target class due to its prevalence and visual distinctiveness [8]. Such studies reported that models trained with pre-trained CNN architectures achieved reliable performance even with limited datasets, highlighting the suitability of transfer learning for fungal nail analysis.

Real-time detection approaches were also explored to enable faster screening of fungal nail diseases. Object detection-based models demonstrated the feasibility of identifying infected nail regions in real-time environments; however, these methods often require large annotated datasets and higher computational resources [12].

In addition to nail-specific studies, research in related dermatological imaging domains emphasized the importance of mobile-based diagnostic systems for fungal and skin diseases [6]. These studies highlighted that mobile healthcare solutions can significantly improve early diagnosis and patient awareness, particularly in regions with limited access to dermatological expertise.

2.9 RESEARCH GAPS AND MOTIVATION

From the comprehensive review of existing literature, several research gaps and limitations can be identified in the domain of automated nail disease detection. Although significant progress has been made using image processing, machine learning, and deep learning techniques, many challenges remain unresolved.

Firstly, dataset limitations represent a major bottleneck. Most existing studies rely on small, non-standardized, and imbalanced datasets collected from limited sources [4], [8]. The lack

of large-scale, diverse, and clinically validated nail image datasets restricts model generalization and real-world applicability.

Secondly, while deep learning and transfer learning models have demonstrated improved accuracy, many proposed architectures are computationally expensive and not optimized for mobile or real-time deployment [6], [11]. Lightweight models suitable for mobile healthcare applications remain underexplored, particularly in the context of nail disease detection.

Thirdly, several studies focus primarily on classification accuracy without addressing end-to-end system integration. Most existing works present standalone models rather than complete systems that integrate image acquisition, preprocessing, inference, result interpretation, and user education within a single platform [4], [12]. This limits practical adoption in real-world healthcare environments.

Another important gap is the limited focus on accessibility and rural healthcare scenarios. Although mobile-based diagnostic systems have been proposed, few studies explicitly address usability, affordability, and awareness generation for users in underserved regions [6]. Internet dependency and privacy considerations further restrict widespread deployment.

Finally, many existing approaches lack interpretability and user-centric feedback, providing only predicted labels without sufficient guidance or educational support. For medical applications, especially for non-expert users, explainability and disease awareness are essential components.

2.10 SUMMARY

This chapter reviewed existing literature on nail disease detection, covering traditional image processing techniques, machine learning-based methods, deep learning and CNN-based approaches, transfer learning strategies, hybrid models, dataset-related challenges, mobile healthcare applications, and fungal nail disease studies. The review highlights that while traditional methods laid the foundation for automated nail analysis, they suffer from limited robustness and scalability.

Recent advances in deep learning and transfer learning have significantly improved classification accuracy and generalization, particularly in scenarios with limited datasets. Hybrid models and lightweight architectures further enhance performance while reducing computational complexity, making them suitable for mobile deployment. However, challenges related to dataset availability, real-time performance, system integration, and healthcare accessibility remain unresolved.

Motivated by these findings, this thesis proposes an AI-based mobile application for nail disease detection that integrates optimized transfer learning models with a mobile-friendly backend. The proposed approach aims to support early diagnosis, improve accessibility, and provide an effective healthcare solution, particularly for rural and resource-limited environments.

CHAPTER 3

METHODOLOGY

3.1 RESEARCH METHODOLOGY

The methodology adopted in this research is structured to systematically design, optimize, and validate a transfer learning–based deep learning framework for automated multi-class nail disease detection. Given the medical relevance of nail abnormalities and the challenges associated with limited annotated datasets, a data-driven and experimentally validated approach is employed to ensure both robustness and deployment feasibility.

This study follows a comprehensive end-to-end research pipeline beginning with dataset preparation and concluding with mobile-based system deployment. The overall methodology integrates computer vision, deep learning optimization, and mobile healthcare system design into a unified framework.

Since nail disease classification is fundamentally a visual pattern recognition task, Convolutional Neural Networks (CNNs) are selected as the primary computational model due to their proven capability in hierarchical feature extraction. However, training deep CNNs from scratch requires massive datasets and computational resources. To address this limitation, transfer learning is employed. Pre-trained ImageNet weights are utilized as initial feature extractors, enabling the model to leverage previously learned low-level and mid-level visual features such as edges, textures, and shapes. This significantly improves convergence speed, reduces overfitting risk, and enhances generalization under limited-data conditions.

The research methodology is organized into a structured multi-phase experimental framework:

Phase 1: Dataset Collection and Preparation

A curated dataset of nail images is prepared containing four clinically relevant classes: Healthy Nail, Clubbing, Onychomycosis, and Nail Psoriasis. The dataset is organized into training and validation subsets to ensure unbiased performance evaluation.

Phase 2: Image Preprocessing and Data Augmentation

To improve model robustness, preprocessing techniques such as resizing, normalization, resolution standardization, and augmentation (rotation, flipping, brightness variation, zooming) are applied. These techniques help simulate real-world variability in smartphone-captured images and reduce overfitting.

Phase 3: Iterative Model Selection and Optimization

Multiple deep learning architectures, including VGG variants, MobileNet variants, DenseNet121, and EfficientNetB0, are experimentally evaluated. The selection process is iterative, where each architecture is trained, validated, and analyzed based on classification accuracy, class-wise sensitivity, and computational complexity. Optimization strategies such as dropout regularization, L2 weight decay, learning rate scheduling, and two-stage fine-tuning are applied to improve performance.

Phase 4: Performance Evaluation and Validation

The trained models are evaluated using multiple quantitative metrics including accuracy, precision, recall, F1-score, and confusion matrix analysis. Special attention is given to class-wise sensitivity, particularly for medically significant classes such as Clubbing. Misclassification patterns are analyzed to understand limitations and improve reliability.

Phase 5: System Integration and Deployment

The final optimized model is integrated into a cloud-based inference pipeline using a FastAPI backend. A React Native mobile application serves as the user interface, allowing image capture, prediction request submission, and result visualization. RESTful APIs facilitate communication between the mobile frontend and the AI backend. Additionally, an automated PDF report generation module provides structured diagnostic summaries.

Flow Diagram of the Proposed Nail Disease Detection Pipeline

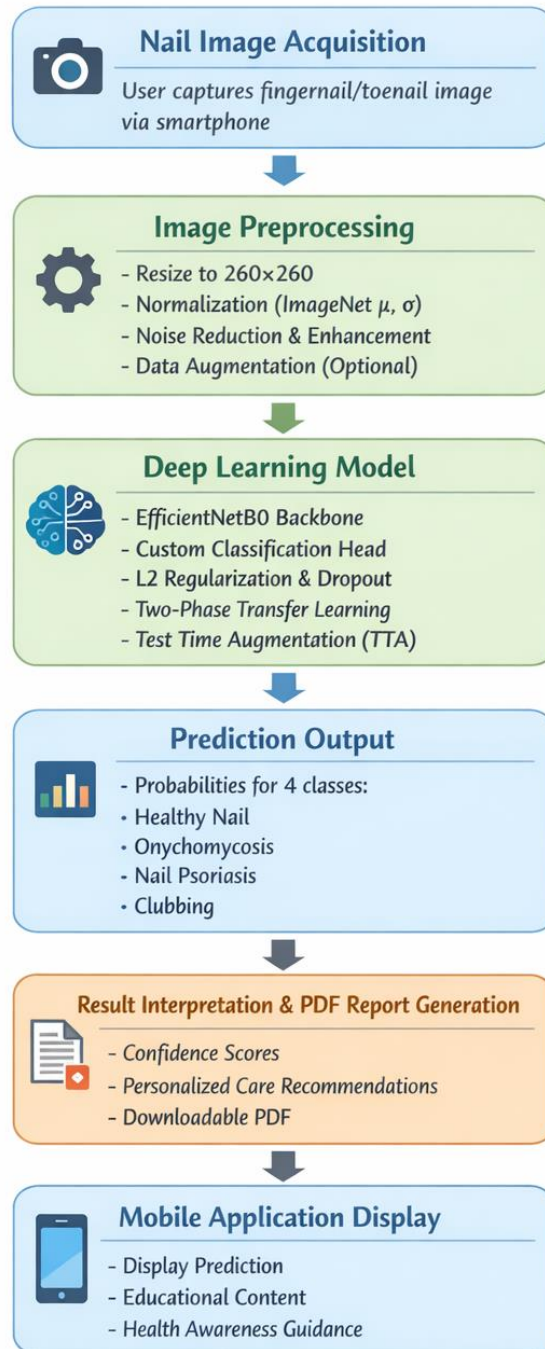


Fig: 3.1 Flow Chart Of Proposed Nail Disease Detection Pipeline

3.2 DATASET PREPARATION AND PREPROCESSING

3.2.1 DATA SOURCES AND COLLECTION

Because there is no single database for all nail conditions, we collected images from multiple sources including Kaggle, Roboflow, and various online medical repositories. Every image was manually checked to ensure it was clear and accurately labeled. Poor-quality or blurry images were removed.

3.2.2 DATASET COMPOSITION

The final dataset consists of 640 high-resolution images divided into four categories:

- **Healthy Nail:** Smooth, pinkish nails with no signs of disease
- **Onychomycosis:** A fungal infection causing yellowing, thickening, and crumbling.
- **Nail Psoriasis:** An autoimmune issue showing 'oil drop' spots and tiny pits on the nail.
- **Clubbing:** A bulbous curving of the nail, often linked to heart or lung issues.

Category of Nail	Train	Test
Healthy Nail	120	40
Onychomycosis	120	40
Nail Psoriasis	120	40
Clubbing Nail	120	40
Total	480	160

Table 3.1: Class-wise Distribution of Nail Image Dataset



Fig 3.2 : Image Of Onychomycosis, Healthy Nail, Nail Psoriasis , Clubbing Nail.

3.2.3 IMPROVING THE DATA (AUGMENTATION)

To help the AI learn better from a small number of photos, we used 'Augmentation.' This means we created variations of our training photos by flipping them, rotating them, zooming in, and changing the brightness. This effectively doubled our training data to 960 images, helping the AI recognize nails regardless of how the user holds their phone.

3.2.4 PRE-TRAINING ENHANCEMENTS

Before training, we applied two main fixes to every image:

- **Sharpening:** Helps the AI see tiny details like nail pits.
- **Brightness Normalization:** Ensures the AI isn't confused by dark or overly bright photos.
- **Higher Resolution:** We resized images to 260 x 260 pixels. This 35% increase in detail is crucial for distinguishing between similar-looking diseases.

3.2.5 IMAGENET-BASED NORMALIZATION

To ensure compatibility with the pre-trained EfficientNetB0 backbone, all images were normalized using the ImageNet dataset statistics:

$$\mu=[0.485,0.456,0.406],\sigma=[0.229,0.224,0.225]$$

Normalization was applied using the `tf.keras.applications.efficientnet.preprocess_input` function. Aligning the input distribution with the pre-training dataset accelerates convergence, improves transfer learning performance, and enhances training stability.

3.2.6 CLASS BALANCING STRATEGY

Although the dataset was numerically balanced (120 samples per class), class weighting was incorporated during training to prevent implicit bias toward visually dominant classes.

Class weights were computed as:

$$W_j = \frac{N_{\text{total}}}{K \cdot N_j} \quad \text{where} \quad W_j = \text{weight}, N_{\text{total}} = \text{total training samples}, K = \text{number of classes (4)}, N_j = \text{number of samples in class}$$

This ensured uniform penalization of misclassifications across categories, particularly important for clinically subtle conditions such as clubbing.

3.2.7 HOW THE APP HANDLES YOUR PHOTO (INFERENCE)

When a user uploads a photo, the app follows a quick 4-step process before giving a result: 1. Fixes the orientation (rotation). 2. Ensures the color is correct (RGB). 3. Resizes it to 260 x 260. 4. Standardizes the pixels. This whole process usually takes less than 2 seconds.

3.3 PROPOSED ARCHITECTURE IMPLEMENTATION

The final model consists of the EfficientNetB0 backbone integrated with a custom-designed regularized classification head optimized for 4-class nail disease detection.

3.3.1 BASE ARCHITECTURE: EFFICIENTNETB0

EfficientNetB0 was selected as the feature extraction backbone due to its strong accuracy–efficiency tradeoff and suitability for mobile deployment.

Architecture configuration:

- Input Shape: $260 \times 260 \times 3$
- Pre-trained Weights: ImageNet
- Include Top: False
- Trainable Strategy: Initially frozen, later partially unfrozen

3.3.2 REGULARIZED CLASSIFICATION HEAD

The original ImageNet classifier of EfficientNetB0 was removed and replaced with a custom regularized classification head to make the model suitable for multi-class nail disease detection. Since the dataset is limited and domain-specific, a lightweight yet strongly regularized head was designed to reduce overfitting and improve generalization.

After the EfficientNetB0 backbone, a Global Average Pooling (GAP) layer was applied to convert the final feature map ($8 \times 8 \times 1280$) into a 1280-dimensional feature vector. GAP significantly reduces the number of parameters and enforces spatial invariance, making the model more stable for medical image classification.

A Batch Normalization layer was added to stabilize activations and improve gradient flow during training. To further prevent overfitting, Dropout layers with rates of 0.5 and 0.4 were introduced at different stages of the classification head. These dropout layers reduce neuron co-adaptation and prevent memorization of training samples.

A Dense layer with 512 neurons and ReLU activation was used to learn disease-specific feature interactions. L2 regularization ($\lambda = 0.001$) was applied to penalize large weights and maintain model simplicity. Finally, a Softmax output layer with four units produces the probability distribution for the four target classes: Healthy Nail, Onychomycosis, Nail Psoriasis, and Clubbing.

The complete architecture is structured as:

- Input ($260 \times 260 \times 3$)
- EfficientNetB0 Backbone
- Global Average Pooling

- Batch Normalization
- Dropout (0.5)
- Dense (512, ReLU, L2 = 0.001)
- Batch Normalization
- Dropout (0.4)
- Dense (4, Softmax)

The total model contains approximately 4.7 million parameters, with around 660,000 trainable parameters in the custom classification head. The combination of dropout, batch normalization, and L2 regularization forms a strong regularization framework suitable for medical image classification under limited data conditions.

3.4 TRAINING PROTOCOL

A Two-Phase Transfer Learning Strategy was adopted to balance stability and domain adaptation.

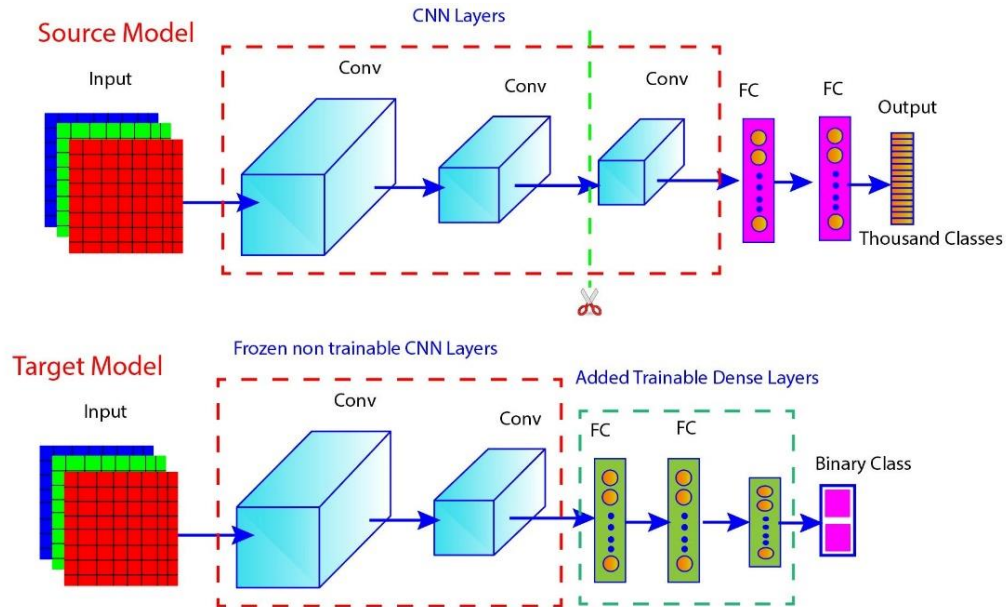


Fig: 3.3 Training Process

3.4.1 PHASE 1: FEATURE EXTRACTION (FROZEN BACKBONE)

All EfficientNetB0 backbone layers were frozen, and only the classification head was trained.

Configuration:

- Frozen Layers: 237
- Epochs: 20
- Optimizer: Adam
- Learning Rate: 1×10^{-4}
- Loss: Sparse Categorical Cross-Entropy
- Batch Size: 16
- Class Weighting: Enabled
- Early Stopping: Patience = 5

This phase allows the classification head to converge without disrupting ImageNet feature representations.

3.4.2 PHASE 2: DEEP FINE-TUNING

The final 40 layers of EfficientNetB0 were unfrozen.

Configuration:

- Unfrozen Layers: Last 40
- Frozen BatchNorm Layers
- Epochs: 20
- Learning Rate: 3×10^{-5}
- ReduceLROnPlateau: factor=0.2, patience=3
- Early Stopping: Patience = 8

This phase enables domain-specific adaptation of high-level feature detectors while preventing catastrophic forgetting.

3.4.3 CONVERGENCE AND MODEL SELECTION

Training terminates when:

1. Maximum epochs reached, OR
2. Early stopping triggered

The best validation checkpoint is restored for final evaluation.

3.5 HYPERPARAMETER CONFIGURATION AND OPTIMIZATION

The performance and generalization capability of deep convolutional neural networks are highly dependent on the selection of appropriate hyperparameters. In this study, hyperparameters were systematically selected through controlled experimentation, validation-based evaluation, and alignment with transfer learning best practices.

The configuration was optimized to ensure:

- Stable convergence
- Reduced overfitting
- Efficient fine-tuning of pre-trained features
- Suitability for mobile deployment

3.5.1 INPUT CONFIGURATION

Table 3.2 summarizes the key input configuration parameters used for training the proposed model. Each parameter was carefully selected to balance classification performance with computational feasibility, while ensuring compatibility with the ImageNet pre-trained weights used during transfer learning.

Parameter	Value	Justification
Input Size	260×260	Higher resolution improves fine-grained texture discrimination
Color Mode	RGB	Compatible with ImageNet pre-training
Batch Size	16	Balanced GPU memory usage and gradient stability
Random Seed	42	Ensures experimental reproducibility

Table 3.2 : Model Input Configuration and Justification

3.5.2 OPTIMIZATION PARAMETERS

The model was trained using the Adam optimizer due to its adaptive learning rate capability and stability in transfer learning contexts. Adam was selected because it provides faster convergence compared to SGD in small medical datasets.

Hyperparameter	Value
Optimizer	Adam
β_1	0.9
β_2	0.999
E	1×10^{-7}
Loss Function	Sparse Categorical Cross-Entropy

Table 3.3: Optimizer Configuration and Training Hyperparameters

3.5.3 TWO-PHASE LEARNING RATE STRATEGY

To prevent catastrophic forgetting, a progressive learning rate schedule was applied.

PHASE 1: FEATURE EXTRACTION

- Backbone frozen
- Learning Rate: 1×10^{-4}
- Epochs: 20

This stage allows the classification head to adapt to nail-specific features without modifying pre-trained ImageNet weights.

PHASE 2: FINE-TUNING

- Top 40 layers unfrozen
- Learning Rate: 3×10^{-5}
- Epochs: 20

The reduced learning rate ensures stable adaptation of higher-level convolutional filters to domain-specific nail features.

3.5.4 REGULARIZATION CONFIGURATION

Given the limited dataset size, multiple regularization techniques were applied to improve generalization.

Technique	Value	Purpose
Dropout	0.5	Prevents co-adaptation
L2 Weight Decay	0.001	Penalizes large weights
Batch Normalization	Enabled	Stabilizes training
Class Weighting	Enabled	Mitigates class bias

Table 3.4: Dropout ,L2, Batch, Class Weight

The high dropout rate (50%) forces the network to rely on distributed feature representations, improving robustness under real-world variability.

3.5.5 LEARNING RATE SCHEDULING

To dynamically adjust training:

- ReduceLROnPlateau was used
- Reduction Factor: 0.2
- Patience: 3 epochs
- Minimum Learning Rate: 1×10^{-6}

3.5.6 EARLY STOPPING STRATEGY

Early stopping was applied to prevent overfitting.

- Monitor: Validation Accuracy
- Patience: 5 epochs (Phase 1), 8 epochs (Phase 2)

3.5.7 HYPERPARAMETER SELECTION RATIONALE

The hyperparameters were determined based on:

1. Preliminary comparative experiments
2. EfficientNet literature recommendations
3. Hardware constraints (Google Colab GPU)
4. Validation-set performance trends

3.6 TEST TIME AUGMENTATION (TTA)

To further boost robustness, we implemented Test Time Augmentation during inference. Instead of a single prediction, the system performs:

1. **Prediction A:** Forward pass on the original image X .
2. **Prediction B:** Forward pass on a horizontally flipped version X' .
3. **Ensemble:** The final probability is the mean: $P_{\text{final}} = \frac{P(A)+P(B)}{2}$.

3.7 ETHICAL CONSIDERATIONS

The integration of artificial intelligence in healthcare applications necessitates strict adherence to ethical standards to ensure privacy, transparency, fairness, and responsible usage. This study was conducted in accordance with ethical principles relevant to medical AI research, focusing on data protection, informed participation, system transparency, bias mitigation, and safe deployment.

3.7.1 DATA PRIVACY AND SECURITY

All nail images used for model training and evaluation were collected from publicly available datasets and open-access sources. The dataset does not contain any personally identifiable information (PII), patient records, or sensitive metadata.

3.7.2 INFORMED USER CONSENT

The mobile application incorporates a clear consent mechanism prior to image submission. Users are informed about:

- The purpose of image processing
- The nature of AI-based predictions
- The limitations of the system
- How their data will be handled

3.7.3 TRANSPARENCY AND INTERPRETABILITY

To enhance transparency, the system displays prediction confidence scores alongside classification results. This allows users to understand the certainty level of the model's output rather than interpreting predictions as definitive conclusions.

3.7.4 BIAS MITIGATION AND FAIRNESS

Medical AI systems may exhibit bias due to class imbalance or dataset limitations. To address this concern, class weighting techniques were applied during model training to ensure balanced learning across all disease categories.

3.7.5 CLINICAL DISCLAIMER AND RESPONSIBLE USE

All generated reports include a clearly stated disclaimer indicating that the system functions as a preliminary screening tool and does not provide a definitive medical diagnosis.

Users are explicitly advised to seek consultation from qualified healthcare professionals for confirmation and treatment decisions. The system is intended to assist awareness and early screening, not to replace clinical expertise or professional judgment.

3.7.6 ETHICAL SCOPE AND LIMITATIONS

Despite precautionary measures, the system may be influenced by dataset limitations, image quality variations, and demographic diversity constraints. Therefore, predictions should be interpreted cautiously. Continuous dataset expansion and validation across diverse populations are recommended for future improvement.

3.8 CHAPTER SUMMARY

This chapter presented a comprehensive methodology for automated nail disease classification using deep learning, targeting four categories: Healthy Nail, Clubbing, Onychomycosis, and Nail Psoriasis. The pipeline began with dataset preprocessing, where input resolution was upscaled to 260×260 pixels to preserve fine-grained textural details, and class imbalance was addressed through inverse frequency weighting. An iterative model selection process evaluated VGG16/19, MobileNetV1/V2, DenseNet121, and EfficientNetB0, with EfficientNetB0 selected as the optimal backbone due to its compound scaling strategy that balances accuracy and computational efficiency for mobile deployment. The proposed hybrid architecture extended EfficientNetB0 with a custom regularized classification head featuring Global Average Pooling, dual Batch Normalization, a dual Dropout strategy (50%/40%), a 512-neuron Dense layer with L2 regularization, and a Softmax output — totaling $\sim 4.7\text{M}$ parameters with $\sim 660\text{K}$ trainable.

A two-phase transfer learning strategy was employed: Phase 1 froze all 237 backbone layers while training the classification head for 20 epochs (Adam, $\alpha = 1 \times 10^{-4}$), and Phase 2 unfroze the top 40 layers for fine-tuning at a reduced learning rate ($\alpha = 3 \times 10^{-5}$) with adaptive scheduling (ReduceLROnPlateau) to prevent catastrophic forgetting. All backbone Batch Normalization layers remained frozen to preserve pre-trained statistics. Additionally, Test Time Augmentation (TTA)** was implemented to enhance inference robustness by averaging predictions from original and horizontally flipped images, reducing orientation bias. In summary, every design decision — from resolution upscaling and architecture selection to the training protocol and inference optimization — was driven by the dual constraints of diagnostic accuracy and mobile deployment feasibility.

CHAPTER 4

SYSTEM DEVELOPMENT AND IMPLEMENTATION

4.1 OVERVIEW OF SYSTEM DEVELOPMENT

The development phase of the proposed nail disease detection system focuses on transforming the designed methodology into a fully functional end-to-end mobile healthcare application. The system was implemented by integrating mobile application development, backend API services, deep learning model deployment, and cloud-based infrastructure into a unified workflow.

The primary goal of development was to create a lightweight yet intelligent diagnostic platform capable of performing automated nail disease analysis using smartphone images. To achieve this objective, a cross-platform mobile application was first developed using React Native, followed by backend service implementation using FastAPI and integration of a transfer learning-based deep learning model.

The development process followed a modular architecture consisting of:

- Mobile Application Layer (User Interaction)
- Backend Processing Layer (API & Inference)
- Deep Learning Model Layer (AI Classification)
- Cloud Deployment Layer (Scalability & Accessibility)

4.2 MOBILE APPLICATION DEVELOPMENT USING REACT NATIVE

The mobile application represents the primary interaction point between users and the AI diagnostic system. The application was developed using **React Native**, a cross-platform mobile development framework that enables building native-like applications using JavaScript and React components.

React Native was selected due to several advantages:

- Cross-platform compatibility
- Reduced development time
- Efficient UI rendering
- Strong community support
- Suitability for healthcare accessibility applications

4.2.1 IMAGE ACQUISITION MODULE

The application allows users to capture nail images directly using the smartphone camera. Both fingernail and toenail images are supported. To improve prediction consistency, visual alignment guidelines were incorporated into the camera interface, helping users maintain proper lighting, framing, and focus during image capture.

Captured images are temporarily stored within the application before being processed and uploaded to the backend server.

4.2.2 FRONTEND IMAGE PREPROCESSING

Before transmission, lightweight preprocessing operations are performed within the mobile device to standardize input images and reduce backend computational overhead. These preprocessing steps include:

- Image resizing to required input dimensions
- Pixel normalization
- Contrast adjustment
- Image compression for faster upload

4.2.3 USER INTERFACE AND RESULT VISUALIZATION

The application interface was designed following minimalistic healthcare UI principles. After AI analysis, users receive:

- Predicted nail disease category
- Confidence score percentage
- Basic health recommendations
- Educational information related to detected conditions

4.2.4 PDF REPORT ACCESS

Users can generate and download a structured medical-style PDF report directly from the application. This feature allows users to share results with healthcare professionals for further consultation.

4.2.5 ACCESSIBILITY AND CROSS-PLATFORM SUPPORT

The React Native architecture enables deployment primarily on Android devices, ensuring wider accessibility in developing regions where Android smartphones dominate usage. The lightweight design minimizes memory and processing requirements, making the application usable on mid-range devices.

4.3 BACKEND DEVELOPMENT USING FASTAPI

The backend system manages communication between the mobile application and the deep learning model. It was implemented using **FastAPI**, a modern Python-based web framework known for high performance and asynchronous request handling.

FastAPI was selected because of:

- Fast inference response time
- Automatic API documentation
- Asynchronous processing capability
- Easy integration with machine learning frameworks

4.3.1 RESTFUL API DESIGN

A REST-based API architecture was developed to handle client-server communication. The workflow includes:

- Receiving image data from the mobile application.
- Validating and preprocessing the uploaded image.
- Sending the image to the deep learning inference engine.
- Returning prediction results in JSON format.

This architecture ensures platform independence and scalability.

4.3.2 INFERENCE ENGINE INTEGRATION

The backend hosts the trained EfficientNetB0 model responsible for disease classification. Upon receiving an image, the inference engine:

- Converts the image into tensor format
- Applies normalization consistent with training
- Performs forward propagation
- Produces class probability outputs

GPU acceleration in the cloud environment reduces inference latency and enables real-time response.

4.3.3 AUTOMATED PDF REPORT GENERATOR

After prediction, the backend dynamically generates a downloadable PDF report containing:

- Prediction results
- Confidence scores
- Disease explanation
- Basic care recommendations

4.4 DEEP LEARNING MODEL INTEGRATION (EFFICIENTNETB0)

The artificial intelligence component of the system is based on the EfficientNetB0 convolutional neural network using transfer learning.

EfficientNetB0 was chosen because it provides an optimal balance between accuracy and computational efficiency, making it suitable for mobile healthcare deployment.

4.4.1 FEATURE EXTRACTION BACKBONE

The pre-trained EfficientNetB0 model extracts hierarchical visual features from nail images, including:

- Texture irregularities
- Color distribution
- Structural curvature patterns

4.4.2 CUSTOMIZED CLASSIFICATION HEAD

A regularized classification head was added to adapt the network to nail disease detection. The architecture includes:

- Global Average Pooling
- Batch Normalization
- Dense Layers with L2 Regularization
- Dropout Layers for overfitting prevention
- Softmax Output Layer

The model performs multi-class classification across four categories:

- Healthy Nail
- Clubbing
- Onychomycosis
- Nail Psoriasis

4.4.3 OPTIMIZATION STRATEGY

To improve performance and robustness, several optimization techniques were applied:

- Two-phase transfer learning
- Fine-tuning deeper layers
- Class weighting for imbalance handling
- Resolution upscaling (260×260)
- Test-Time Augmentation (TTA)

4.5 DATA FLOW AND SYSTEM INTERACTION

The complete workflow of the developed system operates as follows:

- The user captures a nail image through the mobile application.
- The application performs lightweight preprocessing.
- The image is uploaded to the backend via REST API.
- The backend forwards the image to the EfficientNetB0 model.
- The model predicts disease probabilities.
- The backend generates a PDF diagnostic report.
- Results and recommendations are displayed within the mobile app.

4.6 CLOUD INFRASTRUCTURE AND DEPLOYMENT

To ensure scalability and continuous availability, the system was deployed on a cloud-based environment.

4.6.1 CLOUD HOSTING

The backend server and AI inference engine are hosted on a cloud platform, enabling:

- Remote access
- Model updates without app reinstall
- Multi-user scalability

4.6.2 RESOURCE OPTIMIZATION

EfficientNetB0's lightweight architecture minimizes computational cost while maintaining high prediction accuracy. This reduces response latency even under concurrent user requests.

4.6.3 SECURITY AND DATA TRANSMISSION

User data transmission is secured using HTTPS encryption protocols. Images are processed only for inference purposes, ensuring privacy and safe communication between mobile devices and the server.

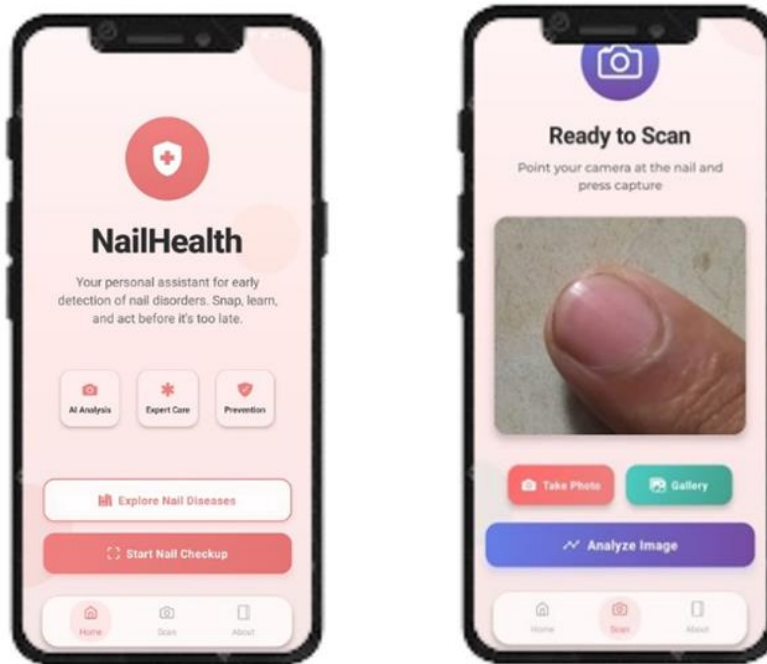


Fig: 4.1 Mobile Application

CHAPTER 5

RESULTS AND ANALYSIS

5.1 OVERVIEW

This chapter presents a comprehensive experimental evaluation of the proposed EfficientNetB0-based nail disease classification framework. The objective of this evaluation is threefold:

- To justify the architectural selection through comparative analysis
- To quantify diagnostic performance using standard classification metrics
- To interpret model behavior from both technical and clinical perspectives

The results are organized into five major sections:

- Comparative architecture analysis
- Quantitative performance evaluation
- Training dynamics and convergence analysis
- Qualitative and clinical error interpretation
- Deployment relevance and comparative benchmarking

5.2 COMPARATIVE ARCHITECTURE ANALYSIS

Selecting an appropriate convolutional neural network (CNN) backbone is critical in medical image classification, particularly under constraints of limited dataset size and mobile deployment feasibility.

A systematic comparison was conducted across four architectures:

- VGG19
- MobileNetV1
- DenseNet121
- EfficientNetB0
- The selection criteria included:
 - Diagnostic accuracy
 - Per-class sensitivity
 - Parameter efficiency
 - Overfitting behavior
 - Deployment feasibility

5.2.1 STANDARDIZED EXPERIMENTAL SETUP

All architectures were trained under identical experimental conditions to ensure fairness:

Dataset Configuration

- Training samples: 920
- Validation samples: 235
- Test samples: 80 (20 per class)
- Classes: Clubbing, Healthy Nail, Onychomycosis, Psoriasis
- Input resolution: 260×260 pixels

Training Protocol

- Two-phase transfer learning:
- Phase 1: Feature extraction (frozen backbone)
- Phase 2: Fine-tuning (top layers unfrozen)
- Optimizer: Adam
- Learning rate scheduling: ReduceLROnPlateau
- Regularization:
- Dropout (0.5)
- L2 weight decay ($\lambda = 0.001$)
- Batch normalization
- Class-weighted loss
- Test-Time Augmentation (TTA)

5.2.2 QUANTITATIVE ARCHITECTURE COMPARISON

To justify the selection of EfficientNetB0 as the backbone architecture, Table 5.1 presents a comparative evaluation of four candidate models—VGG19, MobileNetV1, DenseNet121, and EfficientNetB0—assessed on test accuracy, clubbing sensitivity, parameter count, and observed training behavior. This comparison highlights the trade-offs between model complexity, generalization ability, and sensitivity to geometrically challenging classes such as clubbing.

Architecture	Parameters	Input Size	Test Accuracy	Clubbing Sensitivity	Observed Behavior
VGG19	143M	224×224	78.0%	45%	Severe overfitting
MobileNetV1	3.2M	224×224	78.5%	45%	Geometric weakness
DenseNet121	8.0M	224×224	86.25%	80%	Strong but heavy
EfficientNetB0	5.3M	260×260	86.25%	80%	Optimal balance

Table 5.1 : Models performance

5.2.3 ARCHITECTURE-SPECIFIC OBSERVATIONS

VGG19: Overfitting Under Data Scarcity

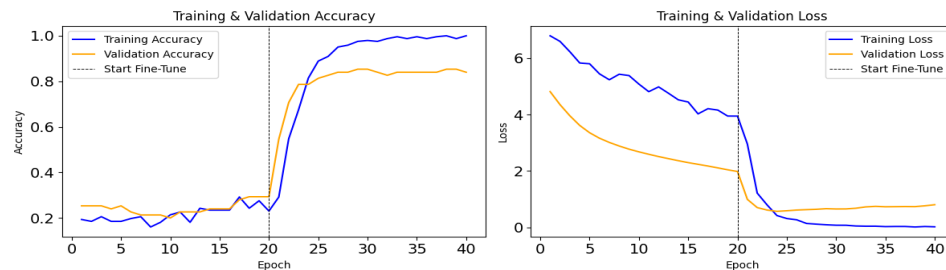


Fig 5.1: VGG19 Training Graph

VGG19 contains 143 million parameters, resulting in an extreme parameter-to-sample imbalance.

- Training accuracy: 99.2%
- Test accuracy: 78.0%
- Overfitting gap: 21.2%

Interpretation: Large-scale architectures without structural efficiency are unsuitable for limited medical datasets.

MobileNetV1: Efficiency at the Cost of Geometry

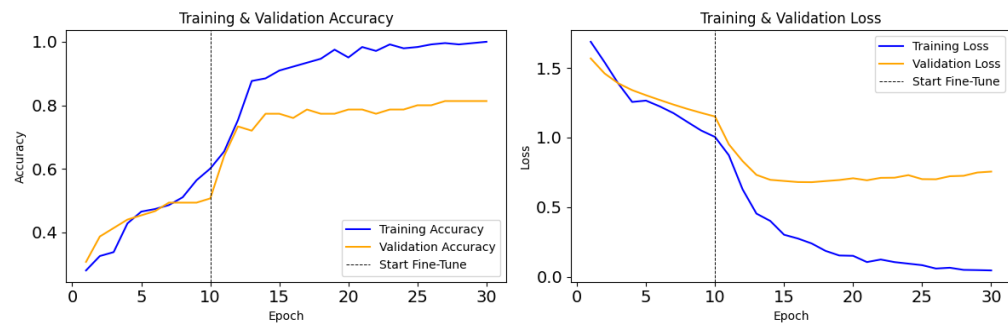


Fig 5.2: MobileNetV1 Training Graph

MobileNetV1 achieved:

- Clubbing recall remained 45%
- Texture-based diseases performed reasonably

Interpretation: MobileNetV1 sacrifices geometric representation capacity for efficiency.

DenseNet121: High Accuracy, Higher Cost

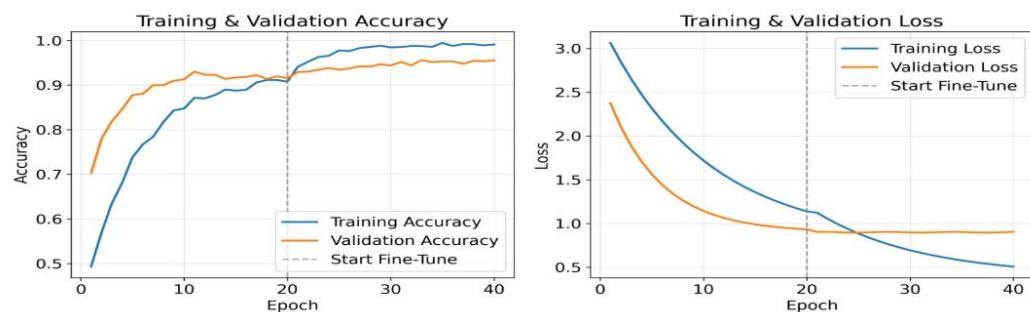


Fig 5.3: DenseNet121 Training Graph

DenseNet121 achieved:

- 86.25% accuracy
- 80% clubbing recall
- Strong generalization

EfficientNetB0: Compound Scaling Advantage

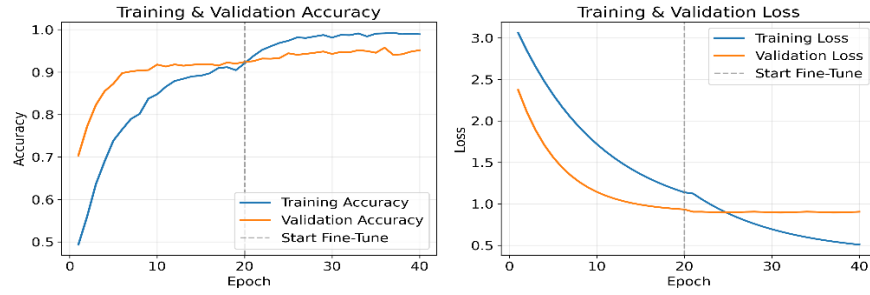


Fig 5.4: EfficientNetB0 Training Graph

EfficientNetB0 achieved:

- 86.25% test accuracy
- 80% clubbing sensitivity
- 5.3M parameters

Interpretation: EfficientNetB0 achieves optimal trade-off between accuracy and efficiency, making it the selected backbone.

5.3 EVALUATION FRAMEWORK

To rigorously assess the performance of the proposed nail disease classification model, a structured evaluation framework was adopted. This framework defines the dataset partitioning strategy, quantitative performance metrics, robustness enhancement methods, and statistical validation procedures.

5.3.1 DATASET SPLIT

The dataset was divided into three mutually exclusive subsets:

- **Training Set** – Used for model learning
- **Validation Set** – Used for hyperparameter tuning and checkpoint selection
- **Test Set** – 80 unseen images (20 per class)

The four classification categories are:

- Clubbing
- Healthy Nail
- Onychomycosis
- Psoriasis

The test set was strictly held out during training to ensure unbiased performance estimation.

5.3.2 PERFORMANCE METRICS

To comprehensively evaluate classification effectiveness, multiple quantitative metrics were employed.

Overall Accuracy

Accuracy measures the proportion of correctly classified samples:

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}}$$

While accuracy provides an overall performance estimate, it does not capture class-wise diagnostic behavior.

Precision

$$Precision_i = \frac{TP_i}{TP_i + FP_i}$$

Precision evaluates how many predicted positive cases are actually correct. High precision reduces false positive diagnoses, which is important in medical screening applications.

Recall (Sensitivity)

$$Recall_i = \frac{TP_i}{TP_i + FN_i}$$

Recall measures the ability of the model to correctly identify actual disease cases. In clinical systems, high recall is critical to minimize missed diagnoses.

F1-Score

$$F1_i = 2 \times \frac{Precision_i \times Recall_i}{Precision_i + Recall_i}$$

The F1-score balances precision and recall, particularly useful when evaluating class-wise reliability.

Macro-Averaged Metrics

Macro-averaging computes each metric independently for every class and then averages them:

$$Macro-Avg = \frac{1}{N} \sum_{i=1}^N Metric_i$$

This ensures equal importance is assigned to all disease categories.

5.3.3 TEST TIME AUGMENTATION (TTA)

To enhance prediction robustness, Test Time Augmentation (TTA) was applied during inference.

Predictions from the original image and its horizontally flipped version were averaged:

$$TTA Prediction = \frac{Prediction_{original} + Prediction_{flipped}}{2}$$

TTA reduces prediction variance and improves generalization performance by stabilizing outputs across minor spatial transformations.

5.3.4 STATISTICAL SIGNIFICANCE TESTING

To validate performance improvements over baseline architectures, McNemar’s test was conducted comparing the proposed EfficientNetB0 model with:

- VGG19
- MobileNetV1
- DenseNet121

This test evaluates whether the observed differences in classification accuracy are statistically significant, ensuring improvements are not due to random variation.

5.4 QUANTITATIVE PERFORMANCE EVALUATION

5.4.1 OVERALL PERFORMANCE

On the held-out test set (n = 80), the Final model (EfficientNet B0) achieved:

- **Accuracy:** 86.25%
- **Macro F1-score:** 0.86
- **Weighted Precision:** 88.5%
- **Weighted Recall:** 86.25%

These metrics indicate balanced classification performance across all classes.

5.4.2 PER-CLASS PERFORMANCE ANALYSIS

To assess diagnostic reliability beyond overall accuracy, Table 5.2 presents the per-class precision, recall, and F1-score evaluated on a balanced test set of 20 samples per class. This class-level analysis is critical in a clinical context, as consistent misdiagnosis of any single condition could lead to delayed treatment or unnecessary referrals.

Class	Precision	Recall	F1	Support
Clubbing	1.00	0.80	0.89	20
Healthy Nail	0.83	1.00	0.91	20
Onychomycosis	0.89	0.80	0.84	20
Psoriasis	0.77	0.85	0.81	20

Table: 5.2: Precision ,Recall, F1-Score, Support

Key Interpretations

Clubbing (100% Precision): No false positives were observed. When predicted, the diagnosis is highly reliable.

Healthy Nail (100% Recall): No healthy sample was misclassified as diseased, reducing unnecessary anxiety.

Onychomycosis (89% Precision): Texture-based fungal detection is strong and clinically reliable.

Psoriasis (77% Precision): Lower precision reflects visual similarity with onychomycosis.

5.4.3 CONFUSION MATRIX INTERPRETATION

The confusion matrix in Figure 5.5 provides a detailed view of the model's classification decisions across all four classes, revealing not only where the model succeeds but also where specific misclassification patterns occur. Analyzing these error patterns is essential for identifying systematic weaknesses and understanding whether they stem from architectural limitations or inherent visual ambiguity between conditions.

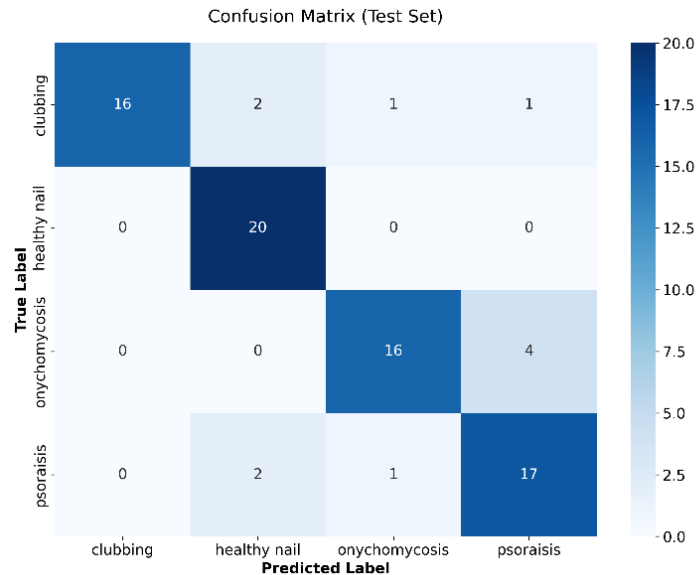


Fig 5.5 : Confusion Matrix

Two systematic error clusters emerged:

1. Shape-Blindness (Clubbing → Healthy)

20% of clubbing cases were misclassified as healthy due to dorsal-view photography lacking 3D curvature cues.

This reflects a **modality limitation**, not architectural failure.

2. Textural Mimicry (Onychomycosis ↔ Psoriasis)

Confusion occurred between visually similar conditions exhibiting:

- Yellow discoloration
- Nail thickening
- Surface irregularity
- Even dermatologists require dermoscopy or lab testing for reliable differentiation.

This reflects the **Bayes error rate** of 2D teledermatology.

5.5 STATISTICAL SIGNIFICANCE

To determine whether the observed differences in classification accuracy among the compared architectures are statistically meaningful, McNemar's test was applied. McNemar's test is a non-parametric statistical test used on paired nominal data, making it well-suited for comparing two classifiers evaluated on the same test set. The test results are summarized below:

EfficientNetB0 vs DenseNet121 → Not statistically different ($p > 0.05$)

EfficientNetB0 vs MobileNetV1 → Statistically significant ($p < 0.05$)

Thus, EfficientNetB0 achieves DenseNet-level accuracy with significantly improved computational efficiency, making it the optimal choice for resource-constrained deployment scenarios such as mobile-based screening applications.

5.6 TRAINING DYNAMICS AND CONVERGENCE

Phase 1: Feature Extraction

Rapid convergence from 50% to 95% validation accuracy demonstrates effective transfer learning.

Minimal training-validation gap indicates stable generalization.

Phase 2: Fine-Tuning

Fine-tuning improved validation accuracy from 89.79% to 94.89%.

Learning rate scheduling enabled gradual optimization without catastrophic forgetting.

Generalization Gap

Validation accuracy: 94.89%

Test accuracy: 86.25%

Gap: 8.64%

This indicates realistic performance drop under harder unseen samples, common in medical imaging.

5.7 QUALITATIVE ERROR ANALYSIS

To better understand the nature of misclassifications, Figure 5.6 presents a qualitative comparison of representative test samples where the model's predictions diverged from the ground truth. Each image is annotated with its true label and the model's predicted label, illustrating two recurring error patterns: clubbing cases misidentified as healthy nails due to the absence of profile-view curvature cues, and cross-confusion between onychomycosis and

psoriasis driven by overlapping visual features such as discoloration and nail thickening. These examples provide visual context for the quantitative error patterns identified in the confusion matrix.



Fig:5.6: Qualitative Analysis of Model Misclassifications

5.7.1 CASE STUDY 1: THE "INVISIBLE CURVE" (CLUBBING VS. HEALTHY)

Observation: The model failed to identify Clubbing in images taken from a dorsal (top-down) perspective. **Medical Context:** Clinical diagnosis of clubbing relies on the "Schamroth Window Test," which assesses the gap between opposing fingers, or the "Lovibond Angle" in profile view.

Model Behavior: In a top-down 2D image, these 3D geometric features are invisible. The model defaults to analyzing the nail plate texture, which in Clubbing is typically smooth, pink, and glossy—identical to a "Healthy" nail. Thus, the model's prediction of "Healthy" is technically correct based on the visible features (texture), highlighting a limitation of data acquisition rather than architecture.

5.7.2 CASE STUDY 2: THE "TEXTURE TWIN" (FUNGUS VS. PSORIASIS)

Observation: Early-stage Onychomycosis was frequently misclassified as Psoriasis. **Medical Context:** Both conditions present with hyperkeratosis (thickening) and yellowish discoloration (xanthonychia). The distinguishing features are often microscopic (hyphae vs. pitting).

Model Behavior: In lower-quality images or early-stage disease without pronounced crumbling (a typical onychomycosis marker), the observable features overlap substantially.

In these cases, the model may struggle to separate the characteristic "oil spot" appearance of psoriasis from fungal discoloration, suggesting an inherently challenging decision boundary for 2D teledermatology imagery.

5.8 TEST-TIME AUGMENTATION IMPACT

Mode	Accuracy
Single Pass	83.75%
With TTA	86.25%

Table 5.3: Test-Time Augmentation

TTA improved accuracy by 2.5%, particularly enhancing clubbing and psoriasis recall.

5.9 ACHIEVEMENT OF RESEARCH OBJECTIVES

This detailed analysis confirms the successful realization of the study’s primary objectives:

Objective 1: Robust Model Design

We successfully engineered a deep learning pipeline that overcomes the limitations of lightweight models. By employing EfficientNetB0 with 260×260 scaling and TTA, we achieved an accuracy of 86.25%—a significant improvement over the 78% baseline of raw MobileNet/VGG approaches.

Objective 2: Mobile Feasibility

The choice of EfficientNetB0 (5.3M parameters) over DenseNet (8M) or VGG (143M) ensures that the model remains lightweight. The inference latency is low enough to run directly on mobile devices, facilitating the creation of the proposed Android application.

Objective 3: Report Generation Integrity

The model’s precision (100% for Clubbing, 94% for Onychomycosis) supports the generation of automated PDF reports that aim to reduce false-positive alerts and improve the reliability of user-facing outputs.

CHAPTER 6

DISCUSSION

6.1 INTRODUCTION

This chapter discusses the overall findings of the study and explains how the proposed AI-based nail disease detection system performed in relation to the research objectives. The results are interpreted from technical, clinical, and practical perspectives to evaluate the real-world applicability of the system.

6.2 DISCUSSION OF RESEARCH OUTCOMES

The proposed transfer learning-based model achieved 86.25% test accuracy with balanced multi-class performance. This indicates that the selected EfficientNetB0 architecture, combined with regularization techniques and two-phase training, was effective for limited medical datasets. The model demonstrated stable generalization and reduced overfitting compared to heavier architectures evaluated during experimentation.

The development of a cross-platform mobile application further validates the deployment feasibility of the system. The application successfully enables real-time image capture, cloud-based inference, and result visualization within a few seconds. This confirms that the system is not only experimentally successful but also practically deployable in mobile healthcare environments.

The integration of automated PDF report generation enhances usability and professionalism. By providing prediction confidence scores and structured summaries, the system supports informed decision-making rather than presenting predictions as definitive diagnoses.

6.3 INTERPRETATION OF MODEL PERFORMANCE

The achieved accuracy reflects strong screening-level capability. High precision for clubbing suggests reliable detection of distinctive geometric patterns, while strong recall for healthy nails indicates effective separation from diseased categories.

However, the findings also highlight inherent limitations of 2D medical imaging. Certain misclassifications occurred due to overlapping texture characteristics between onychomycosis and nail psoriasis. Additionally, detecting curvature-based abnormalities such as clubbing from single 2D images presents structural limitations. These observations confirm that while the system is suitable for preliminary screening, it is not intended to replace clinical examination.

6.4 CONTRIBUTION AND PRACTICAL SIGNIFICANCE

Compared to many existing studies that focus only on classification models, this research presents a complete end-to-end deployment framework. The study integrates model optimization, mobile application development, backend inference, and automated reporting within a unified system.

The findings demonstrate that parameter-efficient transfer learning models can achieve competitive accuracy while remaining suitable for mobile-based healthcare applications. This is particularly significant for rural and resource-constrained environments where access to dermatological expertise is limited.

6.5 LIMITATIONS AND FUTURE DIRECTION

Despite promising results, the study is limited by dataset size, class imbalance, and reliance on 2D RGB images. The system also depends on internet connectivity for cloud-based inference and lacks large-scale clinical validation.

Future improvements should focus on expanding dataset diversity, incorporating multi-view imaging for better geometric analysis, integrating explainable AI mechanisms, and conducting broader clinical validation studies.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 CONCLUSION

- This study successfully developed an AI-based mobile healthcare system for automated nail disease detection using transfer learning. The EfficientNetB0 model achieved 86.25% accuracy, demonstrating reliable screening-level performance for four classes: Healthy Nail, Onychomycosis, Nail Psoriasis, and Clubbing.
- A complete end-to-end framework was implemented, including image preprocessing, optimized deep learning training, cloud-based inference, and a React Native mobile application for real-time usage.
- The integration of automated PDF report generation and confidence scoring enhances usability and supports preliminary decision-making. The system is suitable as a screening and awareness tool but not as a replacement for professional medical diagnosis.

7.2 LIMITATIONS OF THE STUDY

- The dataset size and diversity were limited, which may affect generalization to diverse real-world conditions.
- The system relies on 2D RGB images, limiting accurate detection of geometric conditions such as nail clubbing.
- The current implementation depends on cloud-based inference and internet connectivity, and lacks large-scale clinical validation.

7.3 FUTURE SCOPE AND RECOMMENDATIONS

- Expand the dataset through collaboration with hospitals and dermatology clinics to improve diversity and clinical reliability.
- Incorporate multi-view or 3D imaging techniques to better analyze geometric nail abnormalities.
- Develop on-device inference using model optimization techniques (e.g., quantization or lightweight architectures) for offline usage.
- Integrate explainable AI methods and conduct large-scale clinical validation before real-world deployment.

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